

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1654

Drusilla 100 kV
October 01, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1654. The Interconnection Customer proposes a 141.9 MW solar plus storage facility interconnecting at the Drusilla 100 kV substation in the Illinois area. The request was received on January 24, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 451 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,698,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Taylorville S - Drusilla 100 kV
Contingency: Byron 230-1 kV Autotransformer
Loading: 226.6% (Rating: 129 MVA)
DFAX: 100.0%
Pre-project loading: 116.4%
2. Facility: Taylorville South - Bideford Ip 100 kV
Contingency: Taylorville South 230-1 kV Transformer Bank
Loading: 103.2% (Rating: 290 MVA)
DFAX: 100.0%
Pre-project loading: 54.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Taylorville S - Drusilla 100 kV	Rebuild line with higher-capacity conductor	\$2,256,000	50% (\$1,130,000)
Taylorville South - Bideford Ip 100 kV	Reconductor line	\$1,135,000	50% (\$568,000)
	Total allocated to project		\$1,698,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.